Albert Wang

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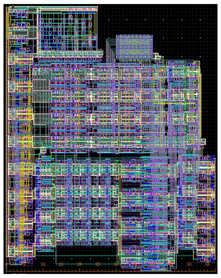
Selected Work

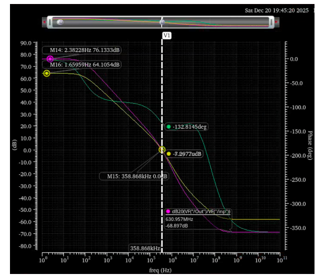


Full-Custom 65nm VLSI Microprocessor — *Fall 2025*

Cadence Virtuoso, Spectre, Calibre DRC/LVS/xRC

Ripple-carry ALU, logarithmic shifter, 8×8 SRAM, and PLA control designed schematic-to-layout in 65nm CMOS. 100 MHz with 88.8% timing margin in 4,100 μm^2; DRC/LVS clean; 876 μW post-layout.

Report (PDF) · GitHub

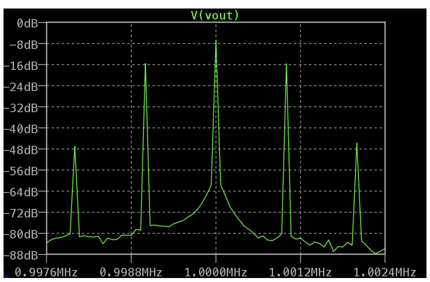


Two-Stage Miller-Compensated OTA — *Fall 2025*

Cadence Virtuoso, Spectre

Two-stage OTA in 0.25 μm CMOS driving an 880 pF load: 76 dB open-loop gain, 47° phase margin, 203 kHz closed-loop bandwidth on a 700 μA budget. Hand calculations matched simulation within 4%; verified across PVT corners.

Report (PDF) · GitHub

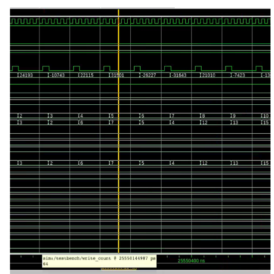


Class D AM Transmitter, 1 MHz — *Spring 2026*

LTspice, discrete BJT design

AM transmitter on a single 6 V supply: RF input buffer, complementary Class D switching pair, and series-LC tank with $Q = 10$ selected from FCC Part 15 emission limits. 84 mW DC, 71% modulation depth, 3.4% THD, 8 dB emission margin.

Report (PDF) · GitHub

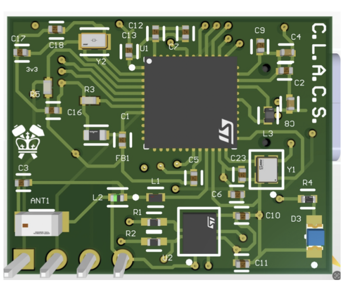


64-Tap FIR Filter with CDC — *Fall 2025*

Verilog, Design Compiler, PrimeTime PX

64-tap 16-bit FIR filter with asynchronous Gray-code FIFO, saturating MAC, and FSM control. Synthesized to IBM 130nm at 100 MHz in 46,551 μm^2; gate-level verified with SDF back-annotation; 3.384 mW at 100% VCD annotation.

Report (PDF) · GitHub



CLACS Smart Pen — *Spring 2026*

Altium, STM32WB55, C firmware

Custom handwriting-capture hardware: designed LiPo/USB-C charging subsystem and MCU schematic, co-developed a two-layer PCB (∼84% area reduction), hand-soldered SMD assembly, and wrote C middleware streaming IMU data over USB-CDC/BLE.

Binder (PDF)

Also Online

Experience

- Hardware Security Intern (PUF/HSM) — eMemory Technology
- Collegiate MEP Intern — PBK Architects / LEAF Engineers

Research

- NSF-REU, Clarkson University — projection-based learning for photonic crystal simulation; 100× speedup; peer-reviewed SPIE Photonics West 2025 publication
- nEXO neutrino experiment, UMass Amherst — SiPM characterization; presented at APS April Meeting 2024

Full documentation — technical reports, schematics, layouts, and source repositories for every project above: albert-wang1010.github.io